SMART INDIA HACKATHON 2026

SMART INDIA HACKATHON 2025

SkillSaathi

From Career Confusion to Family Consensus

- **Problem Statement ID – 26241**
- **Problem Statement Title –** AI-Enabled Career Counselling and Family Decision-Support Platform for Vocational Education
- **Theme –** Smart Education
- **PS Category –** Software
- **Team ID –** [To Be Added]
- **Team Name (Registered on portal) – TeamChargers**



TeamChargers

IDEA TITLE: SkillSaathi — *Beyond Career Advice. Towards Family Consensus.*

SMART INDIA HACKATHON 2025

An AI-powered family counselling platform that bridges student aspirations and parental concerns through evidence-backed vocational guidance.

PROPOSED SOLUTION: FAMILY DECISION INTELLIGENCE WORKFLOW

English + Regional Language | Simple, Family-Friendly Guidance

Student Aspirations
Interests, Aptitude & Ambitions

Parent Concerns
Financial Security, ROI & Stability

AI FAMILY COUNSELLING
Dual-Persona Dialogue Mediation
- Identifies hidden parental objections
- Synthesizes passion with practical ROI

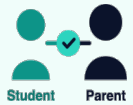
CAREER COMPARISON
Evidence-Based Benchmarking
- Official NSQF/NCVET accredited paths
- Real-world placement & salary data

INFORMED FAMILY DECISION
Consensus & Career Clarity
- Aligned student-parent commitment
- Objective confidence change measured

INNOVATION & UNIQUENESS — WHY SKILLSAATHI?

01. Family Consensus Intelligence ★ FLAGSHIP USP

Identifies student-parent disagreements, addresses parental objections, and tracks changes in family confidence.



Disagreement Detection — Student vs Parent View → **AI Evidence Mediation** — Targeted Objection Fact-Check → **Consensus & Agreement** — Confidence Change Measured

02. Hyperlocal Career Reality Engine

Connects vocational choices to available regional earnings, placement outcomes, and accredited training pathways.



District Cluster — Local ITI & Hubs → **₹ Regional Wages** — Trade Wage Data | **Placement Trends** — Verified Datasets | **NSQF Pathway** — ITI → Poly → B.Voc

03. Explainable Career Trade-off Engine

Compares career options based on family priorities, training costs, earnings, and NSQF progression.



Family Priorities — Budget & Values | **Training Cost** — Fees vs Subsidy | **Time-to-Earn** — 1–2 Yrs vs 4+ Yrs | **NSQF Progression** — Degree Equivalency

04. Parental Resistance Intelligence

Transforms counselling concerns into aggregated regional insights and escalation pathways for scheme administrators.



Unresolved Hesitation — Stigma / Financial Anxiety → **Human Counsellor Routing** — 1-on-1 Escalation Hand-off + **Admin Heatmaps** — District Scheme Insights

TeamChargers

TECHNICAL APPROACH: System Architecture & Implementation

SMART INDIA HACKATHON 2025

An end-to-end technical pipeline translating joint family dialogue into grounded, evidence-backed vocational pathways.

STAGE 01 • React Frontend [Active]

User Interface

- React + TypeScript
- Student & Parent Portals
- Multilingual Interaction

STAGE 02 • FastAPI Services [Planned]

Backend & Data

- FastAPI Architecture
- Family Profiles & Sessions
- Vocational Outcome Datasets

STAGE 03 • Core Innovation [Proposed]

AI Decision Engine

- **Evidence-Grounded Counselling**
- Disagreement Detection
- Career Comparison & Confidence Tracking

STAGE 04 • Action & Governance [Planned]

Outcomes & Analytics

- Family Decision Summary
- Human Counsellor Escalation
- Administrator Dashboard

Official Benchmarks → (NCVET & NCS Data)

DATA GROUNDING ENGINE

Vocational Data: Earnings • Placements • NSQF Pathways

TECHNOLOGY STACK & IMPLEMENTATION SCOPE

FRONTEND [ACTIVE IN CODE]
React, TypeScript, Vite
Modular component library, strict TypeScript contracts, responsive family UI & active mock services.

BACKEND [PLANNED ARCHITECTURE]
FastAPI (Planned)
Pre-architected REST API client (client.ts), typed Pydantic envelopes, family session & profile data models.

AI & DATA [PROPOSED LOGIC]
Grounded AI + Vocational Data
Evidence-grounded decision engine bound to official NCVET qualification packs and NCS wage benchmarks.

OUTPUTS [PLANNED SYSTEM]
Counselling, Referrals, Analytics
Family decision summaries, automated human counsellor escalation routing, and administrative district analytics.

TeamChargers

FEASIBILITY AND VIABILITY

SMART INDIA HACKATHON 2025

Execution Roadmap & Risk Mitigation — Demonstrating practical implementation through a 3-phase progression, grounded data controls, and modular architecture.

SECTION 1: 3-PHASE IMPLEMENTATION ROADMAP *[From Live Prototype to Phased Field Deployment]*

01 PHASE 01 — FOUNDATION [ACTIVE IN CODE]

- **React + TypeScript Frontend:** Operational responsive UI, student & parent intake forms, trade explorer
- **Student-Parent Interface:** Dual-persona guided workflow capturing individual aspirations & concerns
- **Course Discovery Engine:** Structured vocational datasets & local ITI benchmark feeds

02 PHASE 02 — INTELLIGENCE [PLANNED ARCHITECTURE]

- **FastAPI Backend Integration (Planned):** REST API client, session state persistence & data contracts
- **Evidence-Grounded AI Engine (Planned):** RAG counselling bound to NCVET qualification packs & trade criteria
- **Consensus & Confidence Metrics (Planned):** Quantified family agreement scoring & objection tracking

03 PHASE 03 — DEPLOYMENT [FUTURE PILOT ROLLOUT]

- **Vernacular Language Support (Planned):** English + Hindi/regional language testing for inclusive rural access
- **Human Counsellor Escalation (Planned):** Automated triage routing high-resistance families to 1-on-1 counsellors
- **Admin Heatmaps & Pilot Testing (Planned):** District-level aggregate hesitation insights for institutional planners

SECTION 2: CHALLENGES & MITIGATION MATRIX *[Technical Safeguards & Risk Controls]*

Challenge	Mitigation
Limited Verified Data Uncurated regional wage reports	**Evaluation → Official Data** Structured data transitioning to NCVET/NCS APIs
AI Hallucinations Unverified claims or false guarantees	**Grounded RAG + Source Citations** Strict retrieval bounds with source references
Language Barriers Vernacular hesitation among rural parents	**Regional UI + Simplified Layout** Hindi & regional testing with plain terms
Data Privacy Concerns Sensitive family financial ambitions	**Consent Intake + Minimal Retention** Role-based consent with zero session leakage
System Scalability Traffic surges during admission cycles	**Modular Microservices Architecture** Stateless backend design + phased regional rollout

FEASIBILITY PRINCIPLE: Existing Frontend + Modular AI Architecture + Phased Deployment = **Practical Implementation**

TeamChargers

IMPACT AND BENEFITS

Transforming Vocational Decision-Making — Unlocking learner potential by replacing family hesitation with evidence-backed consensus, structured pathways, and institutional alignment.

SMART INDIA HACKATHON 2025

BEFORE SKILLSAATHI *[Current Hesitation Reality]*

- Career Confusion
- Parental Hesitation
- Uncertain Earnings
- Fragmented Guidance



SKILLSAATHI

→

AI Evidence Mediation

WITH SKILLSAATHI *[Evidence-Backed Consensus]*

- Clear Vocational Pathways
- Evidence-Backed Discussions
- Localized Career Insights
- Joint Family Counselling



SECTION 2: MULTI-STAKEHOLDER VALUE CREATION *[Human-Centered Benefits Across the Vocational Skilling Ecosystem]*

01 — STUDENTS
Career Clarity & Opportunity

- Personalized career clarity
- NSQF progression awareness

02 — PARENTS & FAMILIES
Trust & Informed Decisions

- Evidence-backed confidence
- Shared family decisions

03 — TRAINING INSTITUTIONS
Better-Informed Enrolment

- Better-informed applicants
- Potential retention improvement

04 — GOVERNMENT / MSDE
Data-Driven Skilling Decisions

- Regional resistance insights
- Targeted counselling outreach

SECTION 3: LONG-TERM SYSTEMIC MULTIPLIER PATHWAY *[Sustainable Impact from Family Decision to Economic Growth]*

Family Confidence (STEP 1)
Joint dialogue resolves parental hesitation

→

Informed Enrolment (STEP 2)
Learners choose verified vocational tracks

→

Improved Retention (STEP 3)
Family backing supports course completion

→

Skilled Workforce (STEP 4)
Employable, certified youth for industry

" SkillSaathi empowers not just individual learners, but the families shaping their futures. "

TeamChargers

RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2025

Foundational Policy & Academic Frameworks — Grounding SkillSaathi's vocational taxonomies, wage baselines, and family mediation models in verified standards.

SECTION 1: OFFICIAL FRAMEWORKS & RESEARCH REFERENCES *[IEEE Citation Format]*

[1] **National Council for Vocational Education and Training (NCVET), "National Skills Qualification Framework (NSQF),"** *(NCrF-Aligned Qualification Levels & Progression)*
MSDE, Govt. of India, 2023. [Online]. Available: https://ncvet.gov.in/national-skills-qualification-framework/ *[Accessed: Oct. 19, 2026].*

[2] **Ministry of Skill Development and Entrepreneurship (MSDE), "Schemes and National Skill Policy,"** *(National Vocational Mandates & Scheme Norms)*
Government of India. [Online]. Available: https://www.msde.gov.in/ *[Accessed: Oct. 19, 2026].*

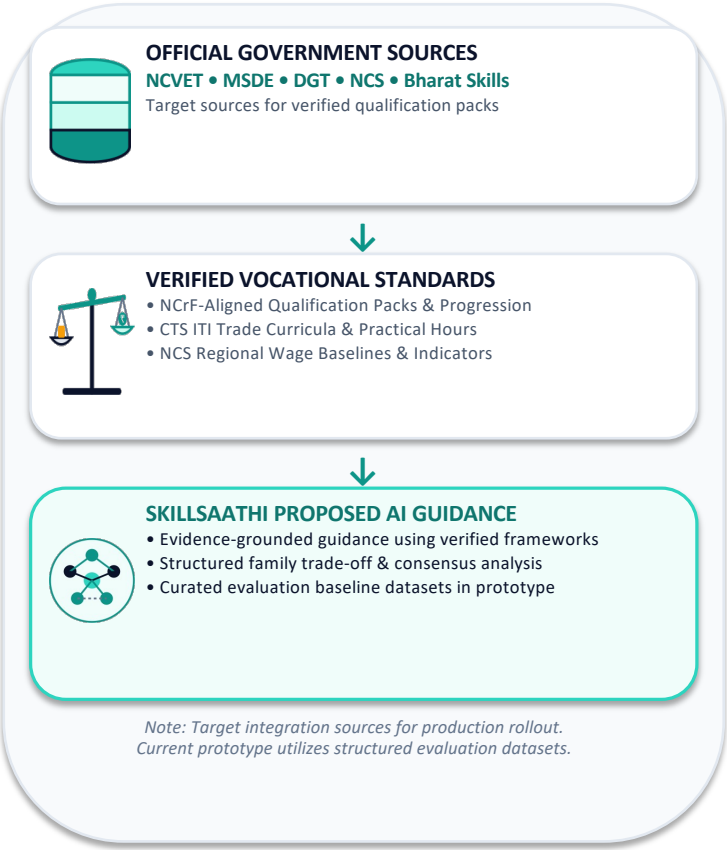
[3] **Directorate General of Training (DGT), "Craftsmen Training Scheme (CTS),"** *(ITI Vocational Trades, Curricula & Duration)*
MSDE, Govt. of India. [Online]. Available: https://dgt.gov.in/en/CTS *[Accessed: Oct. 19, 2026].*

[4] **Directorate General of Employment (DGE), "National Career Service (NCS) Project,"** *(Career Profiles, Sector Demand & Wage Benchmarks)*
Ministry of Labour and Employment, Govt. of India. [Online]. Available: https://www.ncs.gov.in/ *[Accessed: Oct. 19, 2026].*

[5] **Directorate General of Training (DGT), "Bharat Skills Portal,"** *(Trade Learning Repositories & QP/NOS Question Banks)*
MSDE, Govt. of India. [Online]. Available: https://bharatskill.gov.in/ *[Accessed: Oct. 19, 2026].*

[6] **J. Dietrich and B. Kracke, "Career-specific parental behaviors in adolescents' development,"** *(Parental Influence & Family Consensus Research)*
Journal of Vocational Behavior, vol. 75, no. 2, pp. 109–119, Oct. 2009. doi: https://doi.org/10.1016/j.jvb.2009.03.005

SECTION 2: DATA GROUNDING ARCHITECTURE



" SkillSaathi prioritizes verified government frameworks and evidence-grounded recommendations to improve the reliability of vocational career guidance. "